

Dynamic Projection Mapping with 3D Images Using Volumetric Display

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ABSTRACT

Recently, projection mapping has become popular that it is used in various entertainments. In the research field, dynamic projection mapping has been studied to change the appearance of moving and deforming objects. Although it seamlessly connects the real world and the virtual world, complex equipment surrounds the object to achieve advanced projection and it is an unnecessarily visually disturbing factor. Furthermore, projection images are occluded when the target object is grasped and manipulated by hand. In this study, we propose a novel dynamic projection mapping with invisible projection devices with 3D images using a volumetric display and retro-transmissive optics.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

In recent years, spatial augmented reality which visually enhances real space by changing the appearance of the surface of a physical object has come to be used in various exhibitions and entertainments as projection mapping. Furthermore, dynamic projection mapping (DPM) projecting onto an object that can be grasped and manipulated [5] is also being actively studied. However, DPM system such as projectors and cameras are arranged to surround the object in order to achieve advanced projection mapping. This inhibits the presentation of the sense of wonder created by projection mapping. In addition, shadows are generated when an object is grasped and manipulated as in DPM.

In order to resolve them, we have proposed a stealth projection method [3] using mid-air imaging techniques. Stealth projection combines DPM and a projection method [2] that removes the effect of occlusion by applying mid-air imaging techniques. This system reimages light to a symmetrical position in the retro-transmissive optics [4], which allows the projection device such as a projector to be visually hidden. A 3D light source with mirror-symmetrical shape to the target object is placed across the retro-transmissive optics. Projection mapping is achieved when light rays originating from a 3D light source are projected onto a symmetrically arranged target object through the retro-transmissive optics. As the 3D light source required for this system, we have conventionally used a real object called a dummy object projected by a projector. The target object and the dummy object must always maintain a symmetrical position and orientation with 6DOF relative to the retro-transmissive optics. Therefore, the mechanism to synchronize them tends to be complicated.

2 PROPOSED SYSTEM

In this study, we propose a novel stealth projection using a 3D image generated by a volumetric display as a light source (Fig. 1). As mentioned above, stealth projection requires a light source that has

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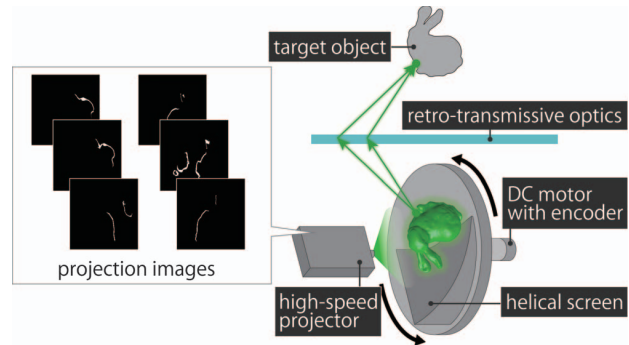


Figure 1: configuration of proposal using volumetric display

the same shape as the target object. In this proposal, we use the volumetric display to generate a light source that is the same shape as the target object, and that can also respond to movement and deformation. As a result, a dummy object and its synchronization mechanism can be removed. This greatly contributes to the simplification of the equipment.

The main feature of the volumetric display is that it can reproduce the same shape as a real object because it forms a light point at a specific position in 3D space. In this research, the main components are a helical screen that can reproduce any depth by sweeping in space through rotation, and a projector that projects images in synchronization with the rotating screen [1]. The screen with a complex helical shape can be easily created using a 3D printer. It is also easy to realize a high-speed projector by using a projector development kit.

3 IMPLEMENTATION

Fig. 2 shows our projection system. The volumetric display consists of a high-speed projector (TI: LightCrafter 4500), a helical screen (PLA plastic, Diameter: 253mm) rotated by a DC motor (maxon: RE40-148867), an encoder (maxon: HEDL5540-110514) and a microcontroller (Arduino) to synchronize the timing of the projector and screen.

By projecting onto a rotating helical screen, a light point of a 3D image is formed in the trajectory of the screen. This light spot is divided into time divisions. By rotating the spiral at a speed fast enough to leave an afterimage in the eye, the light points of each frame are integrated to produce a 3D image of the target object. In order to apply the volumetric display to stealth projection, a 3D image as a light source and the target object are arranged so that their shape and orientation are symmetrical with respect to the retro-transmissive optics. The system is also equipped with a mechanism to achieve the translation and rotation of the target object in one axis. The translation position is measured using an encoder connected to the target by a link mechanism, and the rotation angle is measured using an encoder embedded in the target.

A 3D model of the target object and the spiral is used to generate the projected image. While the spiral is sweeping, the difference of the moment when it intersects the target is extracted and imaged. Projecting this image in synchronization with the rotation angle of

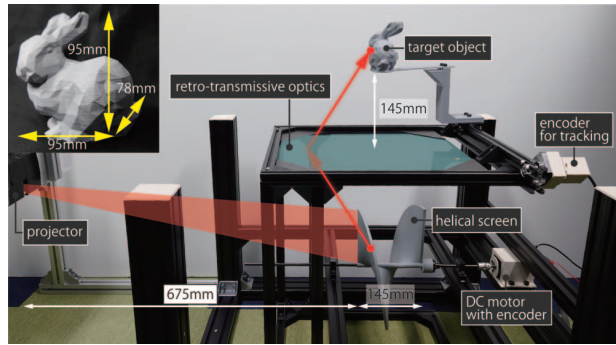


Figure 2: projection system configuration

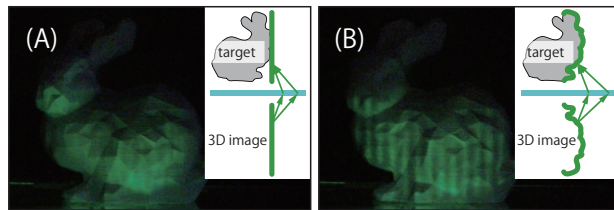


Figure 3: projection with (A) a flat image, (B) a 3D image

the rotating spiral screen generates a part of the 3D light source of the target object. In this implementation, the image is switched and projected every 7.2° rotations of the spiral screen. In order to reduce flickering, five projection images for every 1.44° are combined for generating the projection image for 7.2° .

Our system basically uses a high speed projector. The screen rotates at 900 rpm and the images are projected at 750 fps. However, the high-speed projector is not capable of full color and real-time image input. Therefore, the high-speed projector is used for static projection to verify the basic principle, and a common projector is used to realize DPM. In the latter case, the motion speed of the object and the update rate of the images need to be slowed down. The screen rotates at 60 rpm and the images are projected at 100 fps. Therefore, the evaluation of DPM is basically carried out offline.

4 EXPERIMENT

4.1 Projecting with 3D image

We evaluate the effect of projection with the 3D image using the volumetric display. In this evaluation, the goal is to display a green monochromatic striped texture to the Stanford Bunny. We compare the case of projecting a flat image assuming a standard projector (Fig. 3(A)), and the case of projecting a 3D image reproducing the target object shape by using the proposed methods (Fig. 3(B)). As a result, we can confirm that the proposed method is able to present relatively focused images with visible stripes for the entire range of objects with 3D shapes.

4.2 Projecting on moving object

We evaluate projecting on an object translating and rotating in 1 axis. Since the high-speed projector used in this implementation is not capable of dynamic image updating, a 120Hz projector was used instead. As the update rate decreases, it becomes difficult to observe the projection results with the naked eye. In this study, we conducted a questionnaire survey by recording the projection and presenting the video offline to six male and two female subjects (in their teens and twenties). Fig. 4 shows a part of that video presenting a rotating Stanford Bunny. In the experiment of translation and

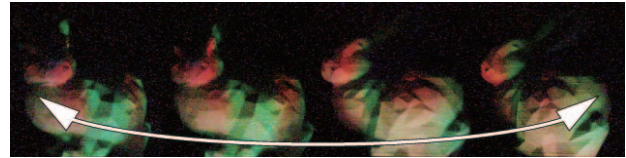


Figure 4: projection result of rotation

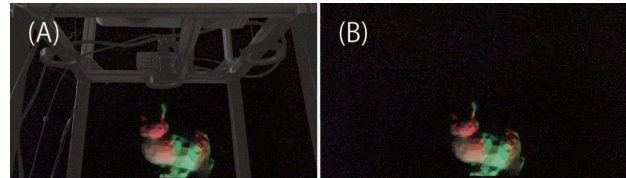


Figure 5: projection environment of (A) conventional DPM, (B) proposed method

rotation, 87.5% of the subjects reported that the texture appeared to follow the motion of the object. Moreover, 71.4% of the subjects in the translating, and 87.5% of the subjects in the rotating reported that the projection made the object look more attractive.

In addition, we asked the subjects to compare the projection mapping under the the equipment used in the conventional DPM and that with our stealth projection (Fig. 5), to see which environment they preferred. The results showed that 87.5% of the subjects in the translating, and 75% of the subjects in the rotating preferred stealth projection environment where the device was invisible. This experiment have suggested that proposal method may help the effect of DPM.

5 CONCLUSION

In this study, we proposed a method of using a 3D image from a volumetric display as a light source for projection onto a 3D object through a retro-transmissive optics. In the future, we intend to improve the brightness of the light source and the resolution in the depth direction, and to realize real-time dynamic image projection.

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